

SehatGuardian AI

Preventive Health Intelligence for India

Team: SehatGuardian Team

Team Leader: [Your Name]

Members: [Member 1, Member 2, Member 3]

College: [College Name, City, State, India]

SDG 3

Good Health & Well-being

The Problem

Core Problem

India has 101M+ diabetics & 220M+ hypertensives — the highest globally.

80% of cases are preventable, yet most Indians lack access to preventive health tools.

Existing solutions are:

- Too expensive (doctor visits cost ₹500-2000)
- Not culturally relevant (Western-centric diet/lifestyle advice)
- Not in local languages
- Focused on treatment, not prevention

Evidence & Impact

101M Indians with diabetes (IDF 2023)

220M+ with hypertension (WHO)

₹4.2L Cr annual economic burden

Only 12% of rural India has access to preventive screening

Affected: Young professionals (25-45), rural populations, families with genetic predisposition — 600M+ Indians at risk

Our Solution

SehatGuardian AI is a free, multilingual, AI-powered preventive health platform built specifically for Indian lifestyles.



Risk Assessment

AI-powered diabetes & BP risk scoring using a 3-layer medical model



Meal Scanner

Analyze Indian meals for glycemic & salt impact with healthier swaps



Lab Decoder

Explain blood reports in simple language anyone can understand



Wellness Guide

Personalized yoga, pranayama & Ayurvedic insights rooted in tradition

Why better? Culturally aware • 6 Indian languages • Free • No diagnosis — only prevention • Integrated traditional wellness

Detailed Solution

3-Layer Risk Model

Population baseline + personal modifiers + interaction effects for accurate risk scoring

Multilingual Support

Hindi, Bengali, Tamil, Telugu, Marathi + English with full UI and AI responses

Institution Dashboard

Campus-wide health analytics, heatmaps, predictive alerts for colleges & corporates

Personalized Action Plans

Daily actionable tasks with gamification — points, streaks, and risk reduction tracking

Traditional Wellness

AI-generated yoga, pranayama & Ayurvedic recommendations based on individual assessment

Privacy First

No diagnosis, no prescriptions. Ethical AI with Supabase RLS, secure edge functions

Scalable: Cloud-native architecture → any Indian language, any institution size, pan-India deployment ready

Technical Architecture



Layer	Technology	Purpose
Frontend	React + TypeScript + Vite + Tailwind CSS + shadcn/ui	Fast, responsive, component-based UI
Backend	Supabase Edge Functions (Deno)	Serverless AI processing & auth
AI Engine	Google Gemini 2.5 Flash (Lovable AI)	Risk analysis, meal scan, lab decode
Database	PostgreSQL (Supabase) + RLS policies	Secure data, user profiles, history
Auth	Supabase Auth (email + password)	Secure user authentication
Cloud/APIs	Lovable Cloud + ElevenLabs TTS	Hosting, voice reading, deployment

Impact & Benefits

Social Impact

- Democratizes preventive health for 600M+ Indians
- 6 Indian languages → inclusive for non-English speakers
- Bridges urban-rural health awareness gap
- Empowers informed health decisions without doctor dependency

Economic Impact

- Prevents ₹50,000-2,00,000/year treatment costs per family
- Reduces healthcare burden on public infrastructure
- Institutional dashboard creates wellness market for colleges/corporates
- Free for individuals — no barrier to entry

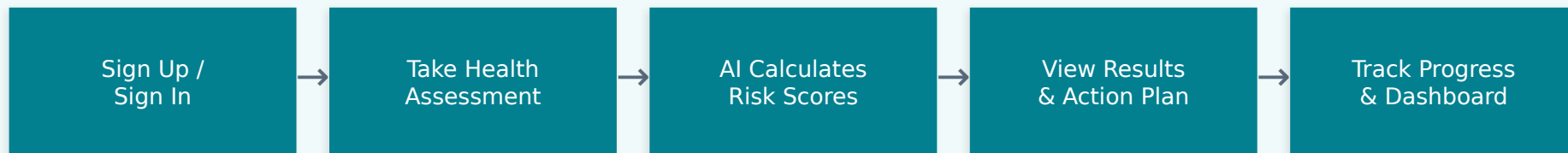
Environmental

- Digital-first: zero paper, zero commute to clinics
- Promotes plant-based traditional diets (lower carbon footprint)
- Reduces unnecessary lab tests through better awareness
- Cloud-native = minimal physical infrastructure

Target Audience: Young professionals (25-45) • College students • Families with genetic risk • Rural populations • Institutions (colleges, corporates)

Live Prototype

📱 Live at: sehat-mitra.lovable.app



Assessment Form

3-step wizard: Basic Info → Lifestyle → Health History. Auto-fills from saved profile.

Risk Results

Visual risk cards with 0-100 scores, confidence levels, "Why These Scores?" with medical reasoning.

Meal Scanner

Photo or text input. AI returns carb/salt impact, diabetes/BP relevance, healthier Indian swaps.

Lab Decoder

Upload or type lab values. AI explains each marker in simple language with status badges.

Wellness Section

Personalized yoga/pranayama and Ayurvedic insights generated based on individual assessment.

Institution Dashboard

Campus heatmap, wellness score, predictive alerts, AI campaign generator for admins.

Research & References

- IDF Diabetes Atlas, 10th Edition (2021) — India: 101 million adults with diabetes
- WHO India NCD Country Profile 2024 — 220M+ hypertension cases
- ICMR-INDIAB Study — Prevalence of diabetes and prediabetes in India
- Lancet Global Health (2023) — Lifestyle interventions for diabetes prevention in South Asia
- WHO Global Report on Traditional & Complementary Medicine (2019)
- AYUSH Ministry of India — Recognition of yoga for NCD prevention
- DASH Diet Research (NIH) — Dietary approaches to stop hypertension
- American Heart Association — Stress and cardiovascular risk guidelines
- Google Gemini AI (2024) — Multimodal AI for healthcare applications
- Supabase & Lovable Cloud — Secure, scalable cloud infrastructure documentation

All references are publicly accessible. Additional research available on request.

SehatGuardian AI

Preventive Health Intelligence for Every Indian

4

AI-Powered
Features

6

Indian
Languages

∞

Lives We Can
Impact

📱 sehat-mitra.lovable.app

Thank You!